



केवल कार्यालयीन उपयोग हेतु
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भारत सरकार GOVERNMENT OF INDIA
रेल मंत्रालय MINISTRY OF RAILWAYS



INTRODUCTORY HANDBOOK ON MICROPROCESSOR CONTROLLED ELECTRIC LOCOMOTIVES (MBFDS)

TARGET GROUP: TRS Maintenance Staff

CAMTECH/ E/ 10-11/ MPC-Loco/ 1.0

July 2010



महाराजपुर, ग्वालियर — 474 005
Maharajpur, GWALIOR - 474 005

*INTRODUCTORY HANDBOOK
ON
MICROPROCESSOR CONTROLLED
ELECTRIC LOCOMOTIVES
(MBFDS)*

QUALITY POLICY

"To develop safe, modern and cost effective Railway Technology complying with Statutory and Regulatory requirements, through excellence in Research, Designs and Standards and Continual improvements in Quality Management System to cater to growing demand of passenger and freight traffic on the railways".

FOREWORD

In the conventional electric locomotives, control is achieved through a large number of relays, switches and auxiliary contacts. This type of control mechanism has mechanical moving parts and a large number of contacts, thus result into malfunction during service.

To overcome these problems microprocessor based control and fault diagnostic system has been developed and is being used in electric locomotives. Since this technology is comparatively new, need has been felt for a handbook to disseminate basic knowledge about the system among the user.

CAMTECH has prepared this handbook which describes important features of the system, various sub-assemblies and input/ output logics etc.

I am sure this document will be very useful for our maintenance staff in electric loco sheds and workshops.

CAMTECH, GWALIOR
DATE : 30TH JULY 2010

S.C. SINGHAL
EXECUTIVE DIRECTOR

PREFACE

Improved technology in the field of microprocessor based digital control systems has made the application of control and fault diagnostic system possible in conventional locomotives.

The microprocessor based control and fault diagnostic system has been developed for conventional electric locomotives including locos provided with static converter. It performs logical control of the locomotive by continuously monitoring various digital/ analog inputs and checks for any abnormality in the operation. It also displays/ announces the fault condition on the display units provided in both cabs of the locomotive. The system is equipped with Real Time Clock (RTC) and a non volatile memory to record the faults in the real time with date, time and fault type.

This handbook on Microprocessor based control and fault diagnostic system has been prepared by CAMTECH with the objective to disseminate basic knowledge of the system.

It is clarified that this handbook does not supersede any existing provisions laid down by RDSO or Railway Board/ Zonal Railways. The handbook is for guidance only and it is not a statutory document.

I am sincerely thankful to all field personnel who helped us in preparing this handbook.

Technological upgradation and learning is a continuous process. Hence feel free to write us for any addition/ modification in this handbook. We shall highly appreciate your contribution in this direction.

CAMTECH, GWALIOR
DATE: 26TH JULY 2010

JAIDEEP GUPTA
DIRECTOR ELECTRICAL

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ISSUE OF CORRECTION SLIPS

The correction slips to be issued in future for this handbook will be numbered as follows :

CAMTECH/E/10-11/MPC-Loco/C.S. # XX date-----

Where “XX” is the serial number of the concerned correction slip (starting from 01 onwards).

CORRECTION SLIPS ISSUED

Sr. No.	Date of issue	Page no. and Item no. modified	Remarks

MICROPROCESSOR BASED CONTROL AND FAULT DIAGNOSTIC SYSTEM OF ELECTRIC LOCOMOTIVES

1.0 INTRODUCTION

There are essentially three main circuits in an electric locomotive viz. power circuit, auxiliary circuit and control circuit. The control circuit is the heart of the locomotive controlling both power as well as auxiliary circuits. The control circuit can be further sub-divided into three parts:

- Traction power control
- Auxiliary machine control
- Light and fan control

There are a large number of relays in control circuit of the locomotive, each relay having multiple contacts. In the conventional locomotives, control is achieved by permutation & combinations of relay contacts, switches and auxiliary contacts of the contactors. As these relays have mechanical moving parts and large number of contacts which creates malfunction of the control system during service.

To overcome these problems microprocessor based control and fault diagnostic system has been developed and is being used in electric locomotives. This fault diagnostic and control system is suitable for all types of electric locomotives including locos provided with static converter.

This system is designed to work with electric locomotives in accordance with RDSO specification no. ELRS/ SPEC/ MPC-FDS/ 001 (REV.2) August 2005.

It performs logical control of the locomotive by continuously monitoring various digital and analog inputs. It also checks for any abnormality in the operation and displays/ announces the fault condition on the display units provided in both cabs of the locomotive.

The system is equipped with Real Time Clock (RTC) and a non volatile memory to record the faults in the real time with date, time and fault type.

Any fault is logged in data packs with event logging of 5 seconds prior and 3 seconds after the occurrence of fault at 1 second interval. The following parameters are recorded in data packs.

- Status of all 128 Digital inputs
- Status of all 80 Digital outputs
- Analog input voltages and currents
- Status of notch position and eliminated relays

2.0 ABBREVIATIONS

CU	Control Unit
SCU	Signal Conditioning Unit
ISCU	Intelligent Signal Conditioning Unit
CSU	Current Sensing Unit
CPU	Central Processing Unit
LED	Light Emitting Diode
LCD	Liquid Crystal Display
USB	Universal serial Bus
IF	Interface
ARNO	Single phase to 3 phase AC converter
SI	Static inverter
TM	Traction motor
RTC	Real time clock
PC	Personal computer

3.0 ADVANTAGES

This system has the following advantages over the conventional control system:

- Elimination of a numbers of relays (Time delay relays, sequential relays, indication relays, no volt relay Q30 and over voltage relay Q 20)
- Elimination of notch indication system
- Enhancement of availability of locomotive
- Reduction in numbers of interlocks required
- Reduction in size of master controller
- Reduction in control wiring
- Fault diagnosis, indication and recording
- Easier trouble shooting as status/ fault messages are displayed to crew in the cabs
- Easier to identify recurring problems by error log and event recording analysis
- Easier and least maintenance
- Improved reliability
- Vigilance control and flexibility

4.0 RELAYS/ FUSES REMOVED

CIRCUITS	RELAYS/ FUSES REMOVED
DJ Control circuit	Q44, Q45, Q118, Q30, QCVAR/QSVM, Q46
Aux circuit	Q100, QTD-105, 106, Q119
Signaling circuit	QV-60 to 64, QVLSOL
Other circuit	Q50,51,52, QD-1,2, QRS, QWC, QF-1,2, Q48, Q49, Q20
Fuses	CCDJ, CCLSA, CCA

5.0 ADDITIONAL SWITCHES PROVIDED

SWITCH	LOCATION	FUNCTION
TSACP	TR panel	For ACP circuit
TSFL	TR panel	For AFL circuit
HPAR / HQ-51	TB panel	For bypassing Q-51 function.
BPQD	Driving Desk	For Sanding during wheel slip.

6.0 SYSTEM SUB-ASSEMBLIES

There are two types of systems mostly in use in electric locomotives. One is of Medha make Ver.2, type MCS 657 and the other one is Stesalit make type FDSC 9648.

These systems comprise of the following sub-assemblies:

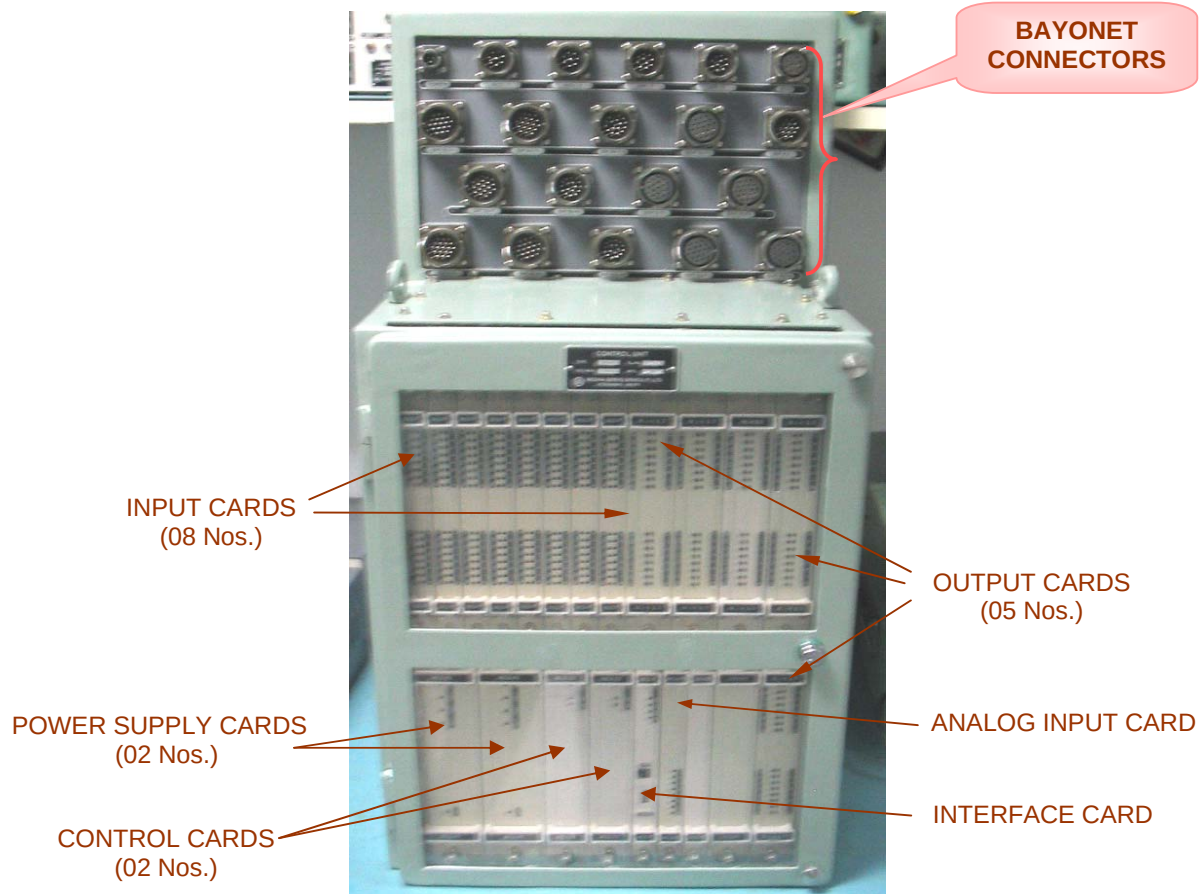
S.No.	Description of Sub-Assembly	MEDHA (MCS 657)	STESALIT (FDSC 9648)
1.	Control unit mounted in AC 2 panel	01 no.	01 no.
2.	Signal conditioning unit mounted above the control unit	01 no.	01 no.
3.	Intelligent signal conditioning unit/ Current sensing unit	02 nos.	02 nos.
4.	Display unit (1 in each cab)	02 nos.	02 nos.
5.	Potential transformer	--	01 Nos.
6.	VCD reset unit (optional)	02 nos.	--

6.1 Control Unit

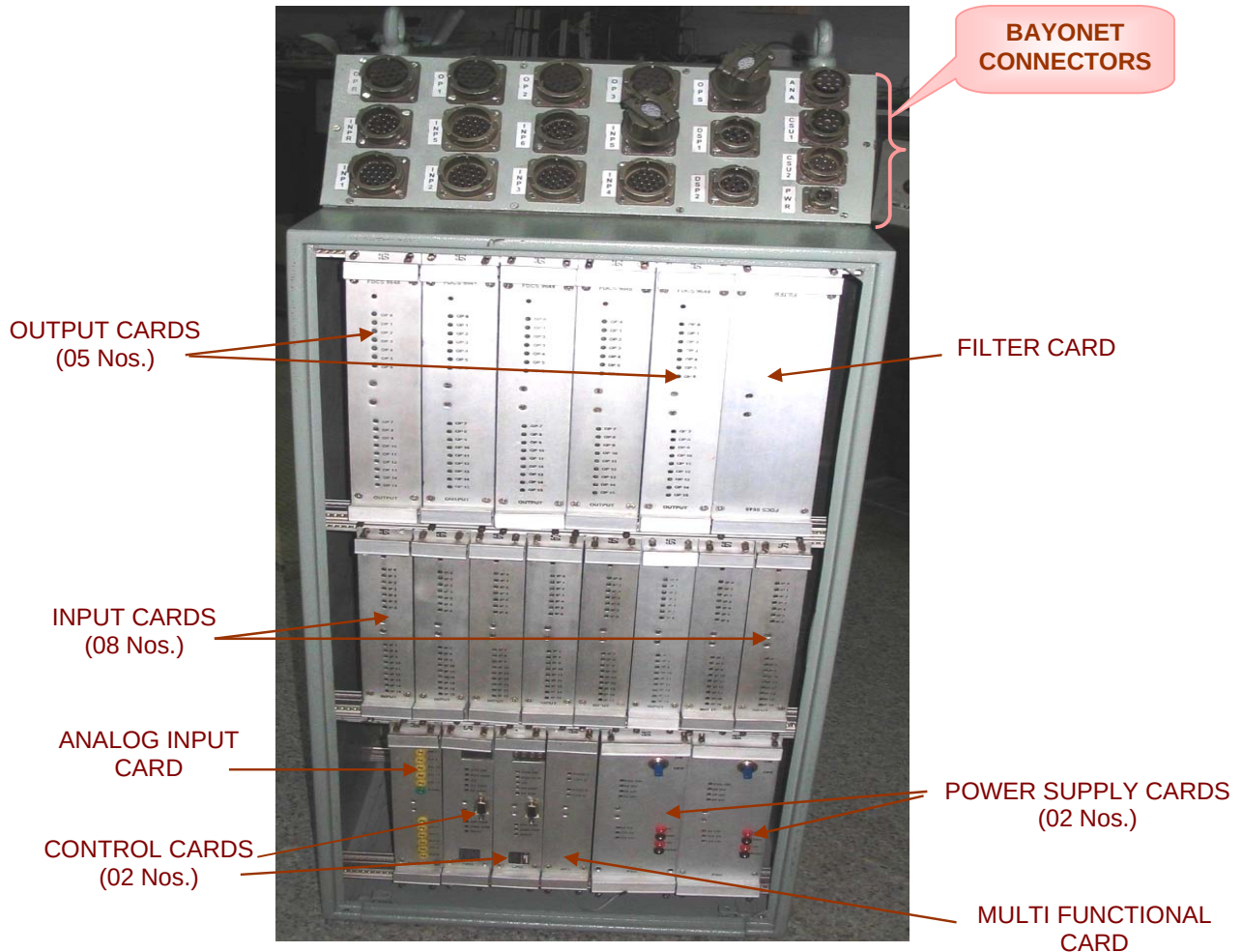
This unit is housed in a powder coated MS enclosure and mounted in AC 2 panel. It consists of the following cards:

S.No.	Control Unit	MEDHA (MCS 657)	STESALIT (FDCS 9648)
1.	Digital input cards	08 nos.	08 nos.
2.	Digital output cards	05 nos.	05 nos.
3.	Analog input card	01 no.	01 no.
4.	Interface card	01 no.	--
5.	Control cards	02 nos.	02 nos.
6.	Power supply cards	02 nos.	02 nos.
7.	Filter card	--	01 Nos.
8.	Multi Function Card	--	01 No.
	Total Cards	19 Nos.	20 Nos.

It also consist necessary Bayonet connectors for all input, outputs, communication and power connections.



MEDHA CONTROL UNIT



STESALIT CONTROL UNIT

6.1.1 Digital Input Card

These cards convert the high voltage input signals of 110 V DC received from the locomotive circuits into isolated low voltage 5 V DC signals which are required for the microprocessor system. All the inputs are protected from surge and reverse polarity. Each card can process 16 input signals. The green colour LEDs provided on the card indicate presence of 110 V DC voltage on that particular channel and the yellow LEDs indicate status of signal read by microprocessor after isolation and signal conditioning. Both the LEDs of a particular channel should be either OFF or ON to indicate the correct functioning of card. Out of 8 digital input cards 1 card is redundant and 1 is spare. These cards are interchangeable.



6.1.2 Digital Output Card

These cards convert low voltage 5 V digital signals which are generated by the microprocessor system to the high voltage signals of 110 V DC to drive relays, contactors, indication lamps etc. in the locomotive. All the outputs are protected from surge, over load, short circuit and reverse polarity. Each card can process 16 output signals. The green colour LEDs provided on the card indicate presence of 110 V DC voltage on that particular channel and the yellow LEDs indicate status of signal given by microprocessor. Both the LEDs of a particular channel should be either OFF or ON to indicate the correct functioning of card. Out of 5 digital output cards 1 card is redundant and 1 is spare. These cards are interchangeable.



6.1.3 Analog Input Card

This card accepts low voltage analog signals given by the signal conditioning unit. These signals are isolated and then taken to the CPU card for processing. LED is provided on each channel for indication and rate of blinking of LED indicates the level of input voltage given to SCU.



6.1.4 Interface Card

This card communicates with both display units. The communication signals are optically isolated from rest of the circuit. The communication signal lines are protected by surge absorbers. This card is also provided with non-volatile memory and real time clock. The fault data is recorded in the non-volatile read write memory with date, time and type of fault. This card is provided with type A USB connector for down loading data to the pen drive and type B miniature connector for configuration of various parameters as well as down loading fault data directly on to PC/laptop. This card is provided with an USB LED to know the status of working with pen drive and blinking of Card Status (CS) LED indicates healthiness of card. Glowing of Watch Dog (WD) LED indicates failure abnormality of CPU card.



6.1.5 Control Card

There are 2 identical Control cards each with 16 bit microcontroller. At a time only one control card is active and other remains in standby mode. This card is provided with Watch Dog Timer Circuit to detect failure of microcontroller or crashing of software, in such condition standby card takes over automatically and system continues to work. Active card is recognized by blinking of LED. It communicates with both intelligent signal conditioning units and the communication signals are again optically isolated from rest of the circuit.



The communication signal lines are protected by surge absorbers. The control card reads analog voltages from Signal Conditioning Unit as well as Configuration information from IF card which can be set by the laptop. It also sends the status of digital inputs, digital outputs, output feedback, eliminated relay status, analog voltages, traction motor currents, diagnosis information, notch position and fault information to the IF card for event logging purpose in case of any fault generated.

6.1.6 Power Supply Card

This module supplies regulated low voltage DC supply to the various cards from locomotive battery. This power supply is protected from reverse polarity, surges, RFI & EMI through suitable devices. This module comprises another card of same type to take over functions of failed card. The following voltages are generated in the power supply card and LED indication is provided for each output.



+9V _I	Isolated DC supply for control circuit
+12V _E	Isolated DC supply for digital output card
+18V _E	Isolated DC supply for signal conditioning unit
-18V _E	Isolated DC supply for signal conditioning unit
+9V _E	Isolated DC supply for communication circuit

6.2 Signal Conditioning Unit

This unit is enclosed in a powder coated MS enclosure. It consists of 5 cards and necessary Bayonet connectors for sending low voltage signals to control unit. Out of these 5 cards, 3 cards are used for AC voltage processing and 2 cards are used for DC voltage processing. These cards are interchangeable with in a group. It accepts locomotive HV inputs of auxiliary supply, ARNO/ SI output, battery charger voltage and TM voltage (of only one Bogie). It has one redundant channel for auxiliary supply.



MEDHA



STESALIT

Analog Inputs

A11	ARNO - U	965
A12	ARNO – V	966
A13	ARNO – W	967
A14	ARNO - M	960
A15	RQ -20	A17
A16	HO- 1	H0 1

6.3 Intelligent Signal Conditioning Unit

This unit is enclosed in a powder coated MS enclosure. It consists of two cards i.e. power supply & control cards along with necessary Bayonet connectors for communication and power connections. Power supply card converts locomotive's 110 V DC to low level working voltage for other cards. Further the control card comprises of three child cards per ISCU for processing of individual TM current. There are two ISCUs on a locomotive to process 6 TM currents; one each mounted in BA1 and BA2 panels. TM shunt voltages are inputs to this unit. Each unit receives 3 TM shunt voltages and processed output is sent to controller unit via communication link. The pick up and drop out currents for auto regression due to wheel slip can be set through configuration settings in the control unit. The power supply input is protected from reverse polarity, surges, RFI & EMI through suitable devices. In Stesalit System TM current sensing unit is used in place of ISCU.



MEDHA



STESALIT (TM current sensing unit)

ISCU – 1

Traction Motor 1 Current
Traction Motor 2 Current
Traction Motor 3 Current

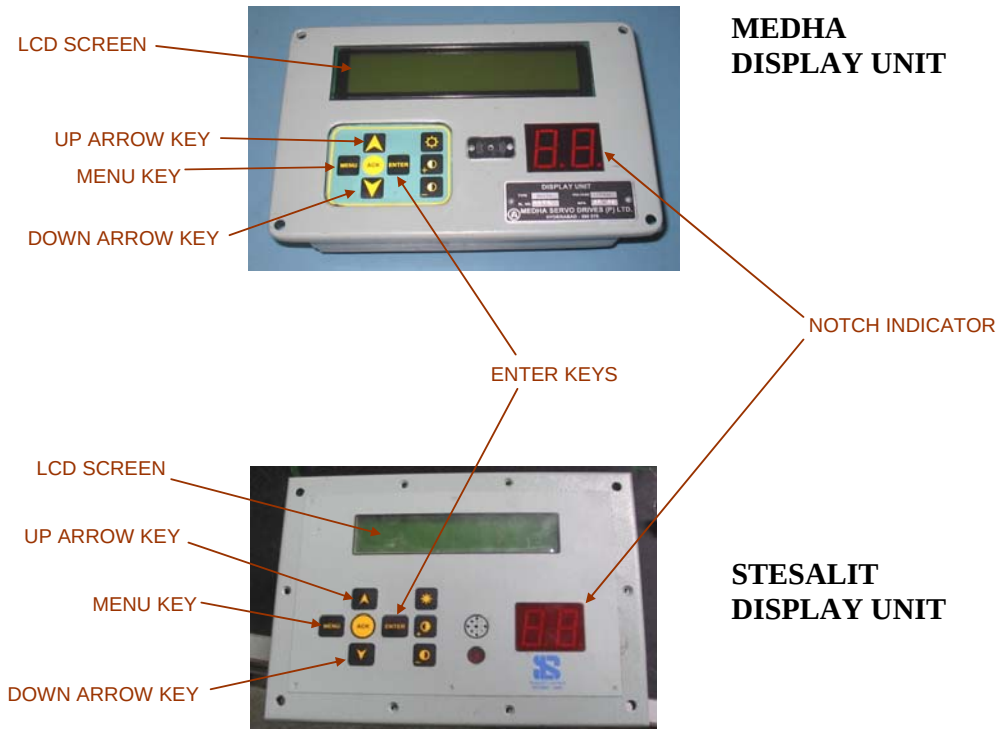
ISCU - 2

Traction Motor 4 Current
Traction Motor 5 Current
Traction Motor 6 Current

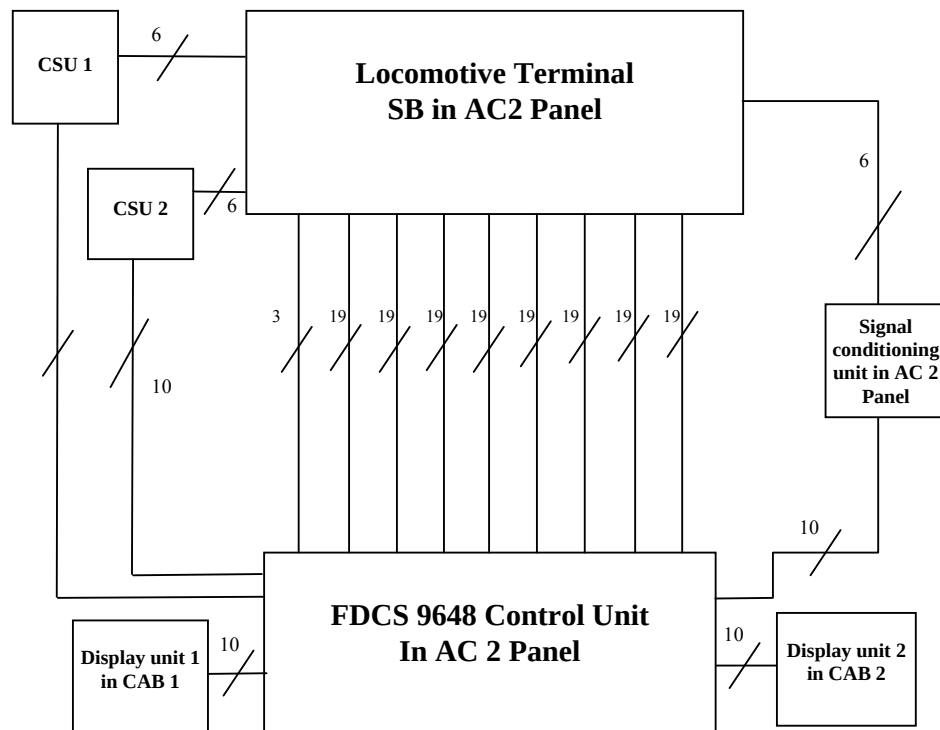
6.4 Display Unit

This unit is enclosed in a powder coated MS enclosure and mounted in each cab of the locomotive. The unit has a 40 character by 4 lines back lit type LCD, a 2 digit 7 segment LED display of one inch height and a 8 keys keyboard with hooter. This unit receives data from control unit. The default LCD screen shows date, time and loco configuration settings. The 7 segment LED display shows current notch no. Any fault is announced with a fault message on LCD with hooter (if configured to announce). The LCD displays the current status, earlier logged faults, isolation conditions, digital/ analog inputs and digital outputs etc. depending on the option selected through the key board. This unit takes 110 V DC from

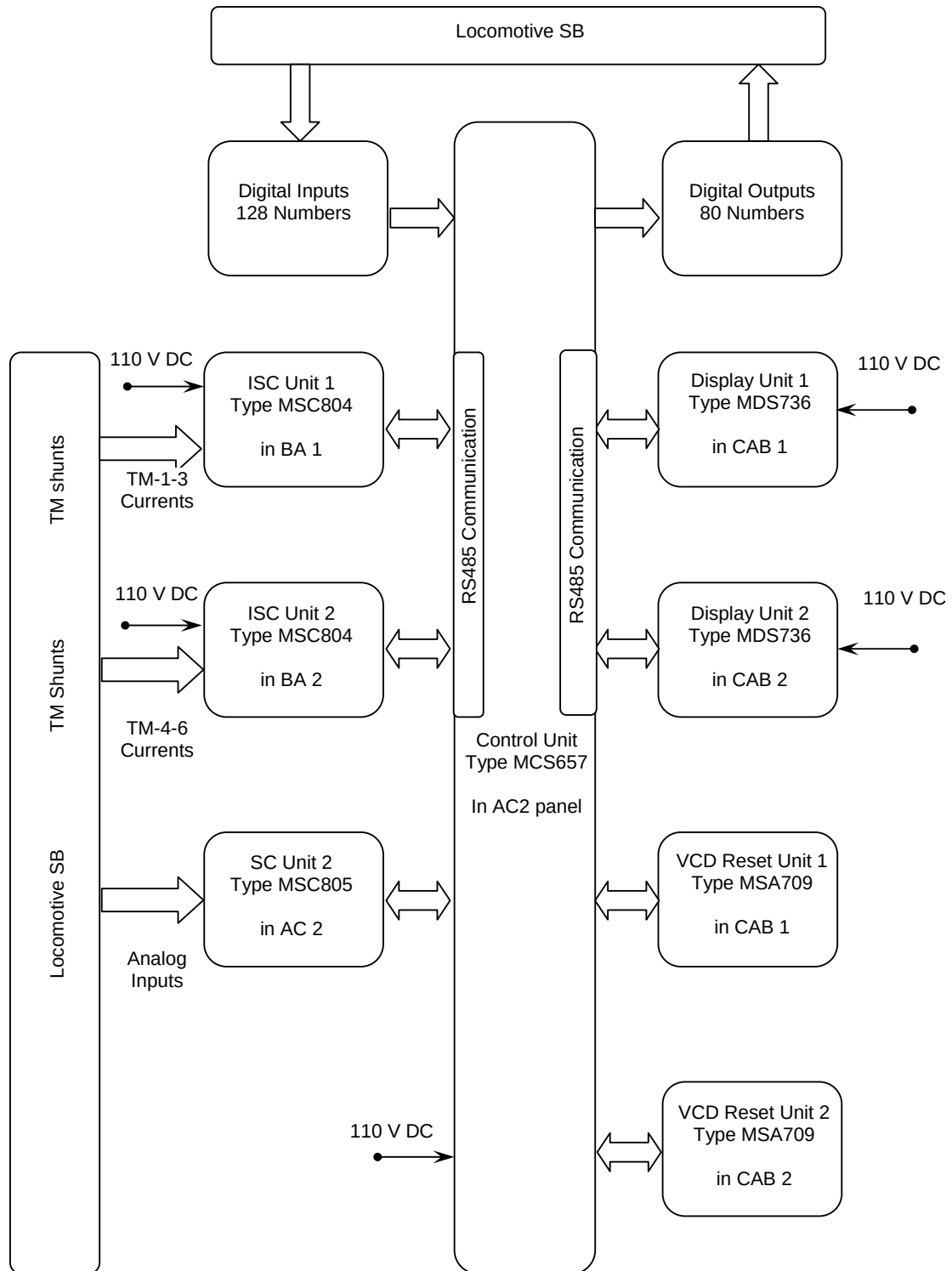
locomotive battery and provides regulated DC low voltages for functioning of display unit. The power supply is protected for reverse polarity, surges, RFI/ EMI, under and over voltages of supply through suitable devices.



7.0 SYSTEM BLOCK DIAGRAM



STESALIT FDCS 9648



MEDHA MCS 657

8.0 TECHNICAL SPECIFICATIONS (MEDHA MCS 657)

8.1 Control Unit

Supply Voltage	60V to 140V DC
Power consumption	25 W approximately
Suitable for locomotives	WAG5, WAG7, WAP4 and WAM4
I/O couplers	Reverse Bayonet connectors
Number of Digital Inputs	128 Nos. with LED indication on fascia
Number of Digital Outputs	80 Nos. with LED indication on fascia
Number of Analog Inputs	5 Nos. provision for 8 nos. with LED indication on fascia
Number of TM Current Inputs	6 Nos.
Weight	55 Kgs approximately

8.2 Display Unit

Supply Voltage	60V to 140V DC
Power consumption	10 W approximately
I/O couplers	Reverse Bayonet connectors
LCD	40 x 4 line alphanumeric with back lit provision
Notch Indicator	Two digit 7 segment indication
Back lit	ON/OFF control
Contrast	Adjustable through keyboard
Keyboard	8 keys of sealed membrane keyboard
Weight	2 Kgs approximately

8.3 Intelligent Signal Conditioning Unit

Supply Voltage	60V to 140V DC
Power consumption	10 W approximately
I/O couplers	Reverse Bayonet connectors
Number of TM Current Inputs	3 Nos.
Weight	3 Kgs approximately

8.4 Signal Conditioning Unit

Supply Voltage	18V DC from Control Unit
Power consumption	10 W approximately
I/O couplers	Reverse Bayonet connectors
Number of Analog voltage channels	5 Nos. (3 for AC voltage and 2 for DC voltage)
Weight	3 Kgs approximately

9.0 PRINCIPLE OF OPERATION

9.1 Control Unit

After the power is switched ON to the Control Unit, Display Unit and ISCU, one of the CPU cards becomes active, it checks for healthiness of various sub systems. Any abnormally found is sent to both Display Units for announcement. The Control Unit reads the Digital and Analog input signals and debounces them through software. Depending on the state of various inputs logical condition of Digital Outputs is worked out based on the predefined logic. Digital outputs are set to high (+ve of Battery supply voltage) or low (Off) state through the Digital output cards. Also the initial status of certain inputs e.g. Reverser is checked at BL key switch ON.

Any abnormality is immediately announced on display Units. The monitoring of inputs and calculation of outputs is done at a very high rate to ensure a prompt action. If any feedback input does not match with the output driving status during running of locomotive, then also an appropriate message is sent to Display Units. If the set limit of TM voltage, OHE voltage, ARNO voltage etc. is crossed, then either Auto regression or DJ tripping is done or any other appropriate action is taken as per the defined logic.

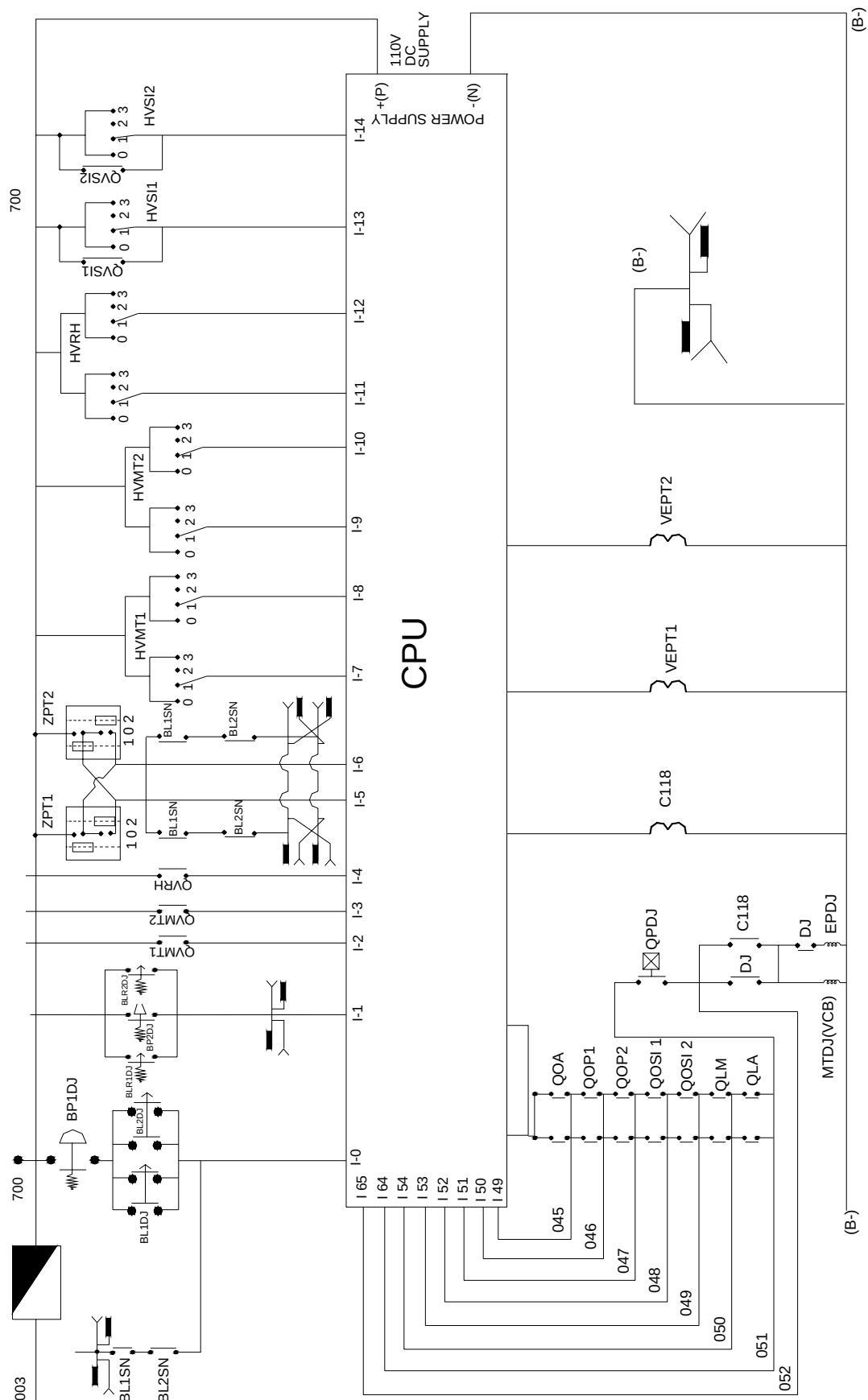
Faults are logged in a Non Volatile Memory with date, time and type of fault and background data. When the fault data is to be downloaded, the USB port of PC/Laptop/ Pen drive is connected to the USB connector of interface card. The downloaded data can be analyzed using the data extraction and analysis software on the PC/Laptop.

For every normal DJ opening with BLDJ, the standby CPU card becomes active and active CPU card becomes standby. If at any stage active CPU card fails, the other healthy CPU card takes over with a message on Display Units and logging of fault. Once a CPU card is found faulty the role change over does not take place. The system works only with the healthy CPU. If both the CPU cards fail, then all the outputs will go into safe mode i.e. de-energized condition (0V).

9.2 Display Unit

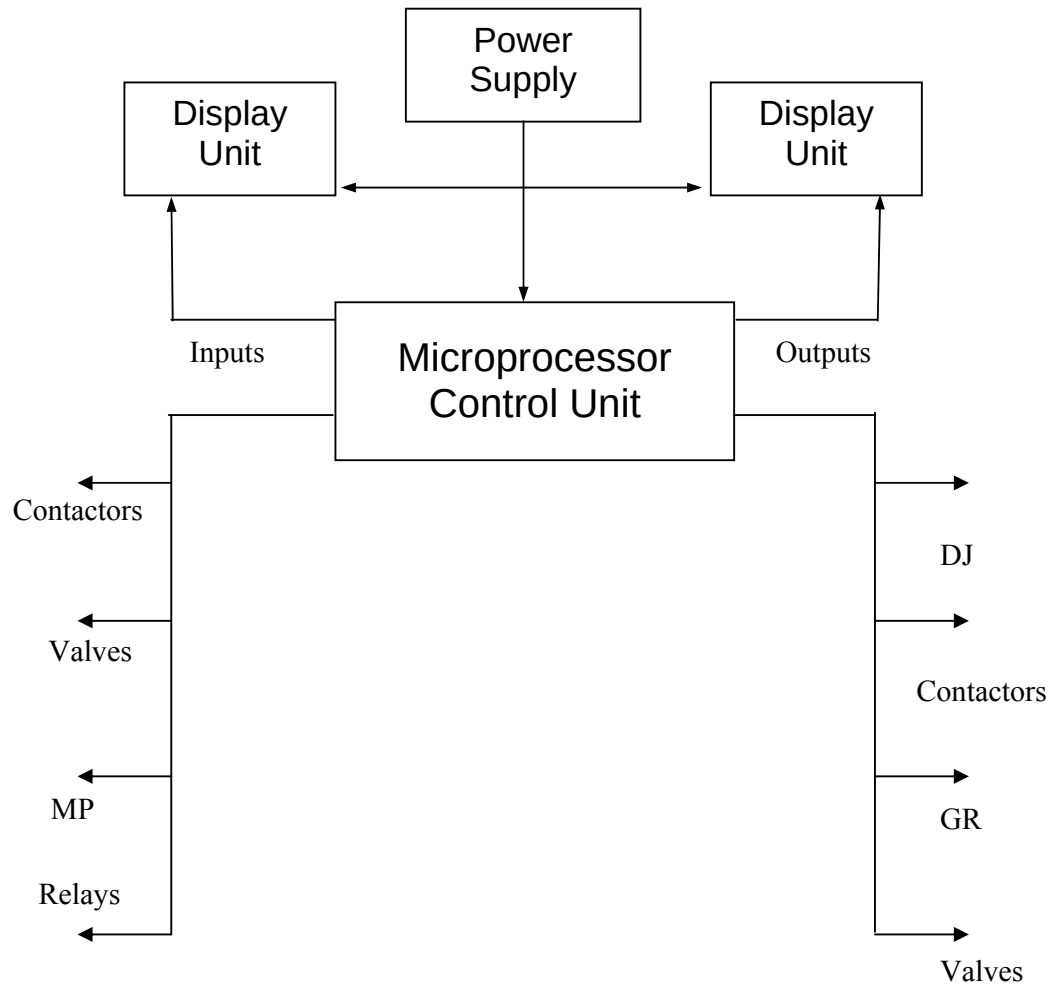
On powering up, each Display Unit checks for communication from Control Unit. If no communication is received within 10 seconds the respective Display announces a communication fail message. The system can continue to work even if one or both Display Units fail. The default screen on the LCD shows date, time and loco configuration which were set. The 2 digit 7 segment LED indicator shows the current notch position. When Control Unit is unable to find out the current notch position e.g. after power up of any of the control unit the tap changer is not at '0', manual operation of GR etc. the display shows "Er".

Any fault message received from control unit is announced by displaying of message on LCD and sounding of hooter (If configure). For messages which need compulsory acknowledgement, the hooter continues to sound till acknowledge key is pressed in the cab where BL key is made 'ON'. For other messages and display in the other cab, the hooter stops sounding after 10 seconds irrespective of the type of message. The keyboard provided on the Display Unit can be used to get the information and current working status of the system. There are 8 membrane keys. These are "MENU", "▲", "▼", "ENTRE", "ACK", Back light ON/OFF, Contrast increase and Contrast decrease.



PANTO & DJ CONTROL CIRCUIT

10.0 OPERATIONAL BLOCK DIAGRAM



11.0 DIGITAL INPUT DETAILS

- ♦ Total No of Digital inputs 128 (I 0 – I 127)
- ♦ Channels per card 16
 - ♦ Main inputs (Cards 1 – 6) I 0 – I 95
 - ♦ Redundant inputs (Card 7) I 96 – I 111
 - ♦ Spare inputs (Card 8) I 112 – I 127

Card No 1

Bayonet Connector: Type - MG06F 22-14 S, Ferrule I0 – I15

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I - 0	BP1DJ/BLDJ	021
B	I – 1	BP2DJ/BLRDJ	024
C	I – 2	QVTM1	025
D	I – 3	QVTM2	026
E	I – 4	QVRH	027
F	I – 5	ZPT1_2	030
G	I – 6	ZPT2_1	029
H	I – 7	HVMT1_1	036
J	I – 8	HVMT1_2	037
K	I – 9	HVMT2_1	038
L	I – 10	HVMT2_2	039
M	I – 11	HVRH_1	040
N	I – 12	HVRH_2	041
P	I – 13	QVSI1/HVSI1	042
R	I – 14	QVSI2/HVSI2	043
S	I - 15	BLVMT	070
T		NC	NC
U		NC	NC
V		BN	BN

Card No 2**Bayonet Connector: Type - MG06F 22-14 SW, Ferrule I 16- I 31**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I – 16	ASMGR (BN)	072
B	I – 17	BLCP/ BLCPD	074
C	I – 18	C101 3FB	075
D	I – 19	GR-0	076
E	I – 20	GR-0 31	077
F	I – 21	QPH/HPH	078
G	I – 22	QVSL1/ HVSL1	079
H	I – 23	C105 FB	061
J	I – 24	ASMGR (ON)	082
K	I – 25	QVSL2/ HVSL2	080
L	I – 26	MP + (R,B)	093
M	I – 27	MPJ (FOR)	091
N	I – 28	J1, J2 (FOR)	095
P	I – 29	MP-(R,B)	096
R	I – 30	MP (+,N,-) R	097
S	I - 31	CTF (RUN)	100
T		NC	NC
U		NC	NC
V		BN	BN

Card No 3**Bayonet Connector: Type - MG06F 22-14 SX, Ferrule I 32-I 47**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I – 32	MPJ (REV)	092
B	I – 33	CTF (BRK)	212
C	I – 34	MP (+,N,-) B	213
D	I – 35	DJ FB	105
E	I – 36	J1, J2 (REV)	107
F	I – 37	ZQWC	121
G	I – 38	MPS (1-4)	123
H	I – 39	MPS (2-4)	124
J	I – 40	MPS (3-4)	125
K	I – 41	MPS4	126
L	I – 42	PVEF	150
M	I – 43	PSA	151
N	I – 44	BPQD	230
P	I – 45	RGEB	155
R	I – 46	SWC	156
S	I – 47	QF & QE	162
T		NC	NC
U		NC	NC
V		BN	BN

Card No 4**Bayonet Connector: Type - MG06F 22-14 SY, Ferrule – I 48 – I 63**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I – 48	BL	142
B	I – 49	QOA/ QSIT	045
C	I – 50	QOP1	046
D	I – 51	QOP2	047
E	I – 52	QRSI1	048
F	I – 53	QRSI2	049
G	I – 54	QLM	050
H	I – 55	BV	122
J	I – 56	C106 FB	062
K	I – 57	ZSMGR	128
L	I – 58	ZSMS	120
M	I – 59	RSI	170
N	I – 60	HMCS	058
P	I – 61	L1 TO L6 FB	028
R	I – 62	HMCS & QD	153
S	I – 63	C118N/C/QCON	018
T		NC	NC
U		NC	NC
V		BN	BN

Card No 5**Bayonet Connector: Type - MG06F 22-14 SZ, Ferrule – I 64 - I 79**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I – 64	QLA FB/SI INT FAULT	019
B	I – 65	QPDJ FB	052
C	I – 66	C107 FB	060
D	I – 67	SI EXT FAULT	068
E	I – 68	C 145 N/O	059
F	I – 69		
G	I – 70	HQ51	200
H	I – 71	CHBA	973
J	I – 72	BL1	149
K	I – 73	SWITI/ DBR	157
L	I – 74		
M	I – 75	RGAF (ACP)	219
N	I – 76	P2	216
P	I – 77	BPT	217
R	I – 78	RGPA/P1 (ACP)	218
S	I – 79	BPSW1/2/ACK (ACP)	203
T		NC	NC
U		NC	NC
V		BN	BN

Card No 6**Bayonet Connector: Type - MG06F 20-29 S, Ferrule – I 80 - I 95**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I – 80	LOCO SEL 1	SEL – 1
B	I – 81	LOCO SEL 2	SEL – 2
C	I – 82	LOCO SEL 3	SEL – 3
D	I – 83	LOCO SEL 4	SEL – 4
E	I – 84	MU FB	
F	I – 85	ES DJ	067
G	I – 86		
H	I – 87		
J	I – 88		
K	I – 89		
L	I – 90		
M	I – 91		
N	I – 92		
P	I – 93		
R	I – 94		
S	I – 95		
T		BN	BN

Card No 7**Bayonet Connector: Type - MG06F 20-29 SW, Ferrule – I 96 – I 111**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I – 13	QVSI1/ HVSI1	042
B	I – 21	QPH/HPH	078
C	I – 22	QVSL1/HVSL1	079
D	I – 25	QVSL2/HVSL2	080
E	I – 49	QOA/ QSIT	045
F	I – 50	QOP1	046
G	I – 51	QOP2	047
H	I – 52	QRSI1	048
J	I – 53	QRSI2	049
K	I – 54	QLM	050
L	I – 64	QLA	051
M	I – 63	C118N/C/QCON	059
N	I – 48	BL	142
P	I – 15	BLVMT	070
R	I – 17	BLCPD	074
S	I – 65	QPDJ FB	052
T		BN	BN

Card No 8**Bayonet Connector: Type - MG06F 20-29 SZ, Ferrule – I 112 - I 127**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	I – 112		
B	I – 113		
C	I – 114		
D	I – 115		
E	I – 116		
F	I – 117		
G	I – 118		
H	I – 119		
J	I – 120		
K	I – 121		
L	I – 122		
M	I – 123	HORN	HORN
N	I – 124	A9	A9
P	I – 125	SA9	SA9
R	I – 126	RESET	RESET
S	I – 127	LOW SPEED	LOW SPEED
T		BN	BN

12.0 DIGITAL OUTPUT DETAILS

- ♦ Total No of Digital outputs 80 (O 0 – O79)
- ♦ Total No of Output cards 5
- ♦ Channels per Card 16
 - ♦ Main Outputs (Cards 1 – 3) O 0 – O 47
 - ♦ Redundant Outputs (Card 4) O 48 – O 63
 - ♦ Spare Outputs (Card 5) O 64 – O 79

Card No.1

Bayonet Connector: Type - MG06F 22-14 P, Ferrule O 0 – O 15

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	O - 0	DJ	044
B	O – 1	C118 BLSI	035
C	O – 2	VEPT1	055
D	O – 3	VEPT2	056
E	O – 4	DJ	044
F	O – 5	C107	083
G	O – 6	C106	084
H	O – 7	C105	085
J	O – 8	C101, C103	086
K	O – 9	VEUL	087
L	O – 10	J1, J2(FOR)	108
M	O – 11	J1, J2 (REV)	109
N	O – 12	CTF (RUN)	111
P	O – 13	CTF (BRK)	112
R	O – 14	VE (UP)	110
S	O - 15	C 145	114
T		BP	BP
U		BP	BP
V		BN	BN

Card No.2**Bayonet Connector: Type - MG06F 22-14 PW, Ferrule O 16 – O 31**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	O – 16	VE (DN)	113
B	O – 17	EVPHGH	115
C	O – 18	SX 1	129
D	O – 19	SX2	130
E	O – 20	SX31	131
F	O – 21	SX32	132
G	O – 22	IP	166
H	O – 23	VESA2	165
J	O – 24	VEF	164
K	O – 25	VESA1	163
L	O – 26	L1, L2, L3	143
M	O – 27	L4, L5, L6	133
N	O – 28	LSDJ (R)	171
P	O – 29	LSCHBA (G)	172
R	O – 30	LSGR (G)	173
S	O – 31	LSB (Y)	174
T		BP	BP
U		BP	BP
V		BN	BN

Card No. 3**Bayonet Connector: Type - MG06F 22-14 PX, Ferrule O 32 – O 47**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	O – 32	LSP (R)	175
B	O – 33	LSRSI (Y)	176
C	O – 34	SX 41	144
D	O – 35	SX 42	145
E	O – 36	LSGROUP (R)	235
F	O – 37	LSOL (Y)	210
G	O – 38	LPAR (R)	179
H	O – 39	SON (ALARM)	177
J	O – 40	Q49/MU	
K	O – 41	B1, B2 (CPA)	231
L	O – 42	LSDBR (Y)	234
M	O – 43	LSFL	232
N	O – 44	QFL	236
P	O – 45	FL LP	
R	O – 46	C102	
S	O – 47	C108	
T		BP	BP
U		BP	BP
V		BN	BN

Card No.4**Bayonet Connector: Type - MG06F 22-14 PY, Ferrule O 48 – O 63**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	O – 48	DJ	044
B	O – 49	C118/BLSI	035
C	O – 50	VEPT 1	055
D	O – 51	VEPT 2	056
E	O – 52	DJ	044
F	O – 53	C 107	083
G	O – 54	C 106	084
H	O – 55	C 105	085
J	O – 56	C101, C 103	086
K	O – 57	VE (DN)	113
L	O – 58	J1, J2 (FOR)	108
M	O – 59	J1, J2 (REV)	109
N	O – 60	CTF (RUN)	111
P	O – 61	L1, L2, L3	143
R	O – 62	VE UP	110
S	O – 63	L4, L5, L6	133
T		BP	BP
U		BP	BP
V		BN	BN

Card No. 5**Bayonet Connector: Type - MG06F 22-14 PZ, Ferrule O 64 – O 79**

Pin No	Signal		Ferrule on Core
	Legend	Name	
A	O – 64		
B	O – 65		
C	O – 66		
D	O – 67		
E	O – 68		
F	O – 69		
G	O – 70		
H	O – 71		
J	O – 72		
K	O – 73		
L	O – 74		
M	O – 75	WORKING	WORKING
N	O – 76	FAULT	FAULT
P	O – 77	EM VALVE	EM VALVE
R	O – 78	BUZZER	BUZZER
S	O – 79	WARNING	WARNING
T		BP	BP
U		BP	BP
V		BN	BN

13.0 OUTPUT TO INPUT LOGIC

O0 & O4 (DJ closing)	I0=1(BLDJ-on, Wire no-021), I1=1(BLRDJ-on, Wire no-024) I5,I6 or i55=1(ZPT1-on,WireNo-029, or ZPT2-on Wire no-030 or BV-on Wire no-122), I16=1(ASMGR IL ½ Between notches-off, Wire no-072), I19=1(ASMGR IL 41/42-on,Wire no-076),I20=1(ASMGR IL 61/62 – on Wire no-077), I23=0(C105 Feedback-off, Wire no-061), I24=1(ASMGR on notch-on, Wire no-082), I48=1(BL1/2-on,Wireno-142), I72=1(BL1-on,Wire no-149 For working from Cab1), I49=1(QOA-off, Wire no-045), I50=1(QOP1-off,Wire no-046), I51=1(QOP2-off,Wire no-047), I52=1(QRSI1-off,Wire no-048), I53=1(QRSI2-off,Wire no-49), I54=1(QLM-off, Wire no-050), I56=0(C106FB-off,Wire no-062), I63=1(C118-off,Wire no 059/018), I64=1(for QLA-off & for SI int. flt.I64=0), I65=1(QPDJ-on, Wire no-052), I66=0(C107FB-off Wire no-068 & Analogue Input Voltage-Aux. Voltage-215Volt & Arno Voltage-157Volt
O1 (C118 Closing)	IO=1,I1=1,I5orI6orI55=1I19=1,I48=1,I63=1 I0-BLDJ-on wire no-021 I1-BLRDJ-on wire no024 I5,I6&I55-Any one on wire no-029,030&122 respectively I19-ASMGR-on wire no-076 I48-BL-on wire no-142 I63-C118-off wire no-059/018 I35-DJ feedback-off wireno-155
O2 (VEPT1)	I6=1 ZPT-on wire no029&I55=0 BV-off wire no122
O3 (VEPT2)	I5=1 ZPT-on wire no-030& I55=0 BV-off wire no-122
O5 (C107 Coil)	I15=1BLVMT-on wire no-070, I11=1HVRH(1or3) wire no-040, I35=1 DJ feed back wire no105, I68=0 C107 feed back wireno-080

O6 (C106 Coil)	I15=1BLVMT-onwire no070, I9=1HVMT2-on (1or3) wire no-038, I56=0(C106) feedback wire no-062
O7 (C105Coil)	I15=1BLVMT-onwire no-070, I7=1HVMT1-on (1or3) wire no-036, I23=0(C106) feedback wire no-081
O8 (C101&C103) Coil	I19=1BLCP&RGCP or BLCPD on wire no-074, I35=1DJ feedback wire no-105, I18=0 C101,102,103 feedback wireno-075, I63-1QCON/CII8 on wire no-059/018
O9 (VEULCoil)	O8=0-1 wire no086 or I18=1-0 wire no-075
O10 (J1&J2 Up Coil)	I48=1BLon wire no-142, I19=1 SMGR on-0 wire no- 076, I27=1(MPJ1-F or MPJ2-R) wire no-091
O11 (J1&J2 Dn Coil)	I48=1Blon wire no-142, I19=1 SMGR on-0 wire no- 076, I32=1(MPJ1-R or MPJ2F) wire no-092
O12 (CTF1,2,3Up Traction)	I48=1Blon wire no-142, I30=1MP on Traction wire no-097
O13 (CTF1,2,3Dn Braking)	I48=1Blon wire no-142, I34=1MP on Braking wire no-213
O14 (VE1 UP Coil)	I48=1BL-on wire no-142, I35=1DJ feedback wire no105, I20=1SMGR on 0to31 wire no-077, I26=1(MP on+ Tr or Br)wire no-093, O31=0 LSB-off wire no-174, I55=1 RGEB on wire no -155, I16=0SMGR Dn-off wire no-113
O15 (C145 Coil)	I35=1DJ feedback wire no-105, I34=1MP on Braking wire no-213, I33=1CTF1,2,3 on Braking feedback wire no-212, I27&I28=1or I32&I36=1Reversar F/R, I60=1Both HMCS on-1wire no-058, I61=0L1-6 feedback wire no-028, I47=1QE feedback wire no-162, I73=1QVRF on/SMGR up to5 notch wire no-159, I46=1SWC on wire no -156, I57=1ZSMGR on wire no-128, I58=1ZSMS on 1 (On MP)wire no-120

O16 (SMGR DnVE2)	In running- I35=1DJ feedback wire no-105, I19=0SMGR not on 0 wire no-076, I29=1MP on-(Tr or Br) wire no-096, O31=1LSB off wire no-174, I45=0RGEB off wire no-155, On other condition(Auto Regression), I45=0RGEB-off wire no-155, I35=0 DJ feedback wire no-105, O38=1(ACP actuated), QD Operated(Current difference between TM2&TM3 Or TM4&TM5, TM Over voltage ,Tm Voltage Above 810 Volt
O17 (EVPHGR Coil)	SMGR above 5notches
O18 (Shunting Contactors1)	I38=1MPS1-onwire no-123, SMGR above20 notch
O19 (Shunting Contactor2)	I39=1MPS2-onwire no-124, SMGR above20 notch
O20 (S13,S23,S33)	I40=1MPS3-onwire no-125, SMGR above20 notch, I37-1BPQWC-on(MPJ-FOR)
O21 (S43,53,63)	I40=1MPS3-onwire no-125, SMGR above20 notch, I37-1BPQWC-on(MPJ-REV)
O22 (IP Valve)	I35=1, O22 Will be off when I35=0 In Braking Mode
O23 (VESA-2)	I36=1J1J2 on Rev I48=1 or QD on
O24(VEF)	I35=1 I42=1PVEF-onwireno-150&BP more than 3.5Kg, In Braking Mode O24 Will not be High
O25 (VESA-!)	I28=1J1J2 on FORI48=1 or QD on
O26 (L1,L2,L3)	I35=1DJ feedback-on wire no 105, I7=1 HVMT1 on1/3 wire no -036, I30=1 MP on run wire no097, O31=0 LSB-off
O27 (L4,L5,L6)	I35=1DJ feedback-on wire no 105, I9=1 HVMT2 on1/3 wire no -038, I30=1 MP on run wire no097, O31=0 LSB-off
O28 (LSDJ Off)	I35=1

O29 (LSCHBA Off)	I71=1CHBA Voltage above-85 Volt, I63=1QCON-on for STC loco
O30 (LSGR Off)	I19=0 SMGR not on 0
O31 (LSB Off)	(1) Running mode I35=1DJ feedback-on, Either I27 with I28=1 Or I32with I36=1(J1&J2 Both are in for or Rev), I68=0C145FB-off, I31=1 CTF on run, I30=0=0or1, I34=0 MP on braking (2) Braking Mode I35=1DJ feedback-on, Either I27 with I28=1 Or I32with I36=1(J1&J2 Both are in for or Rev), I68=1C145FB-on, I31=0CTF on braking, I30=0=0or1, I34=1MP on braking, I46=1SWC-on, I33=1 CTF on braking, I47=1QE-off, I60=1 Both HMCS on1, I61=0 Line contactor-open, I68=1 C145 close, I73=I QVRF on, I48=1 BL on
O32 (LSP)	I77=1 BPT-on, QD1orQD2 operated (Current difference more than 170Amp(WAG7), 125Amp(WAP4) inTM2&TM3 or TM4&TM5
O33 (LSRSI)	I48=1 BL on, I59=1 RSI micro switch on
O34 (S14,24,34) Shunting4	I41=1MPS3-onwire no125
O35 (S44,54,64) Shunting4	I41=1MPS3-onwire no125
O38 (LED ACP)	I76=1
O39 (SON)	TM voltage above 810 Volt
O44 (QFL)	I76=1 But I79 should not go high
O46 (C102)(081)	5 second after O8=1
O47 (C108)	O15=1 C145 close &O31=0

14.0 DO's

1. Check all connections at input/ output connectors & ensure they are not interchanged. This will lead to system malfunction and may lead to accident.
2. Simulation test is mandatory after every change of cables/components, new program loading etc. to avoid any malfunction incident on line.
3. Use trained/ confident staff only to attend these locos as the trial and error methods are not at all acceptable and may lead to major fire or such fatal things.
4. Ensure dust proof covering for all the system components as it will avoid malfunction and extend useful life of the equipment.
5. Always ensure vibration free fitment of the equipment.
6. Download the fault data and analyze carefully to take corrective action.

15.0 DON'Ts

1. Never try to bypass any input/ output cards by using external cables without properly studying the repercussions after such bypassing.
2. Never try to modify the main configuration settings of the system without proper knowledge of the software/ hardware.
3. Never switch OFF the battery supply while the system displaying “Busy with USB communication”.
4. Never do any Hammering, Welding, Gas cutting in the vicinity of the CPU as it may lead to malfunctioning or permanently damage the system.

16.0 GENERAL GUIDELINES

1. Encourage all concerned staff to always follow systematic approach while working on the sophisticated equipment.
2. Make available all related information like trouble shooting and maintenance manuals
3. Use proper tools & handling equipment.
4. Record all activities carried on a particular system.
5. Maintain good understanding with firms service personnel to give and take the information as and when required.

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2. Operation, Trouble Shooting and Maintenance manual of Microprocessor Based Controlled and Fault Diagnostic System Type MCS 657 of Medha Servo Drives Pvt. Ltd., Hyderabad.
3. Field study and Literature collected from various electric loco sheds/ workshops.
4. Presentations given by participants from various Electric Loco Sheds during seminar conducted on 16TH July 2010 at IRCAMTECH/ Gwalior.

OUR OBJECTIVE

To upgrade maintenance technologies and methodologies and achieve improvement in productivity, performance of all Railway assets and manpower which inter-alia would cover reliability, availability, utilisation and efficiency.

If you have any suggestions and any specific Comments please write to us.

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